AIM:

To write a program to find the LU Decomposition of a matrix.

Equipments Required:

1. Hardware – PCs
2. Anaconda – Python 3.7 Installation / Moodle-Code Runner

Algorithm

```
1. Start the program
2.Import the necessary libraries(numpy,scipy.linalg)
3.Define the matrix using numpy
4.Use lu(),lu_solve(),lu_factor() to get the solutions
5.End the program
```

Program:

(i) To find the L and U matrix

```
Program to find the L and U matrix.
Developed by: vishwajith p
RegisterNumber: 212225220122

import os
os.environ["OPENBLAS_NUM_THREADS"]="1"
import numpy as np
from scipy.linalg import lu
A=np.array(eval(input()))
P,L,U=lu(A)
print(L)
print(U)
```

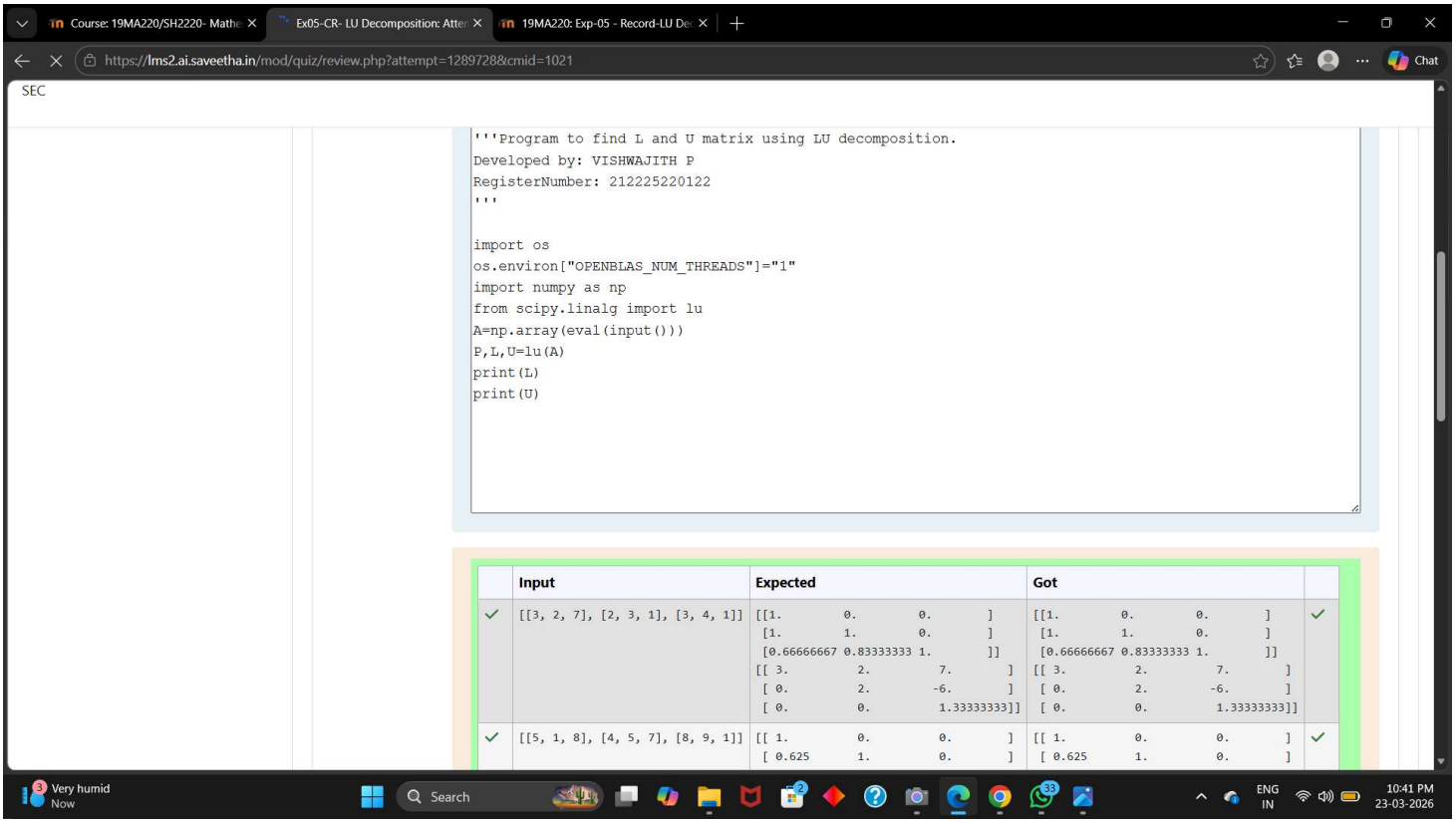
(ii) To find the LU Decomposition of a matrix

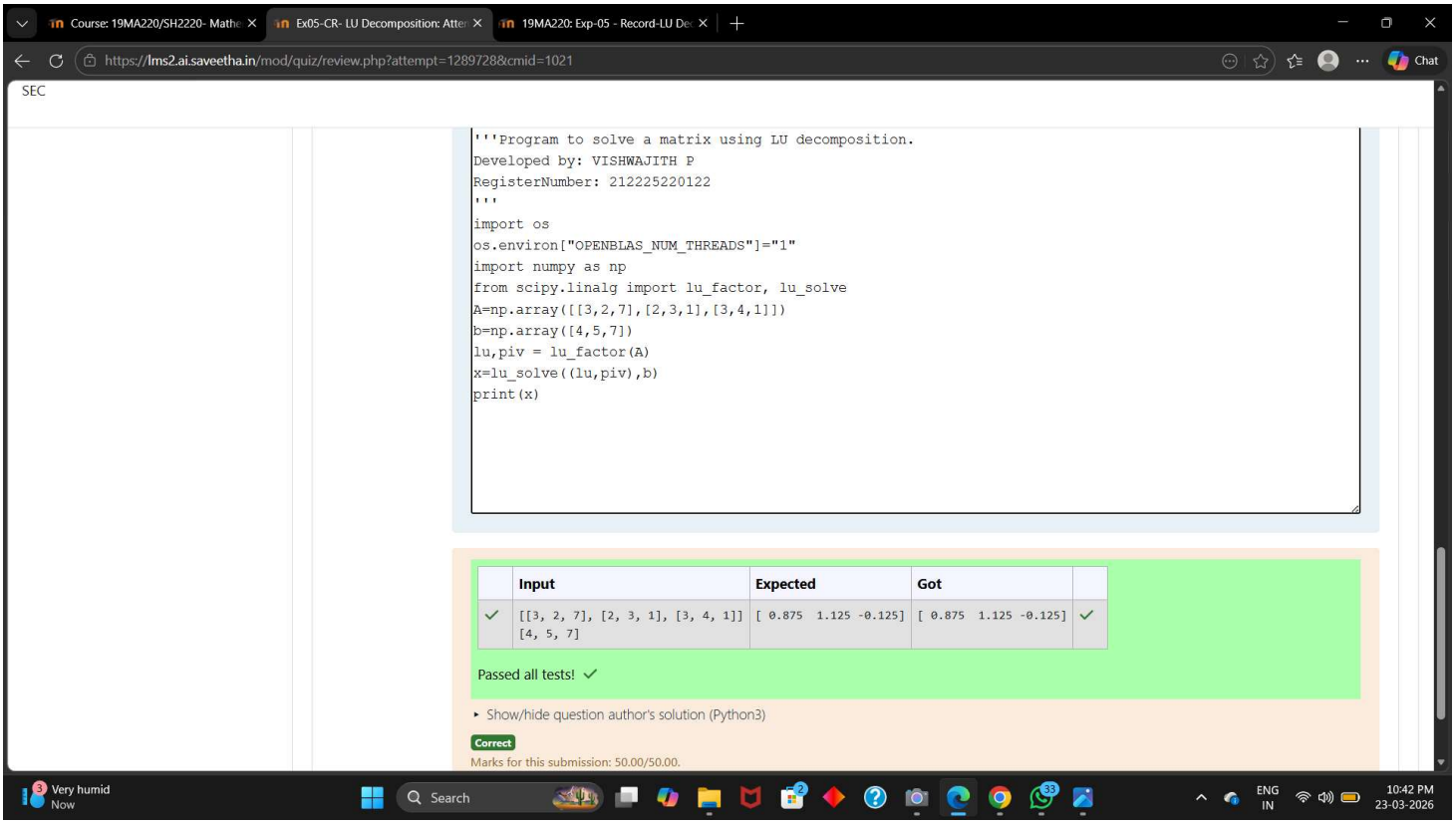
```
Program to find the LU Decomposition of a matrix.
Developed by: vishwajith p
```

```
RegisterNumber: 212225220122

import os
os.environ["OPENBLAS_NUM_THREADS"]="1"
import numpy as np
from scipy.linalg import lu_factor, lu_solve
A=np.array([[3,2,7],[2,3,1],[3,4,1]])
b=np.array([4,5,7])
lu,piv = lu_factor(A)
x=lu_solve((lu,piv),b)
print(x)
```

Output:





Result:

Thus the program to find the LU Decomposition of a matrix is written and verified using python programming.